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Pacemaker Implantation Associated with Tricuspid Repair in the Setting of Mitral Valve Surgery: Insights from a Cardiothoracic Surgical Trials Network Randomized Trial

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*A complete list of members of the trial is provided in the Supplementary Appendix.

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Abstract

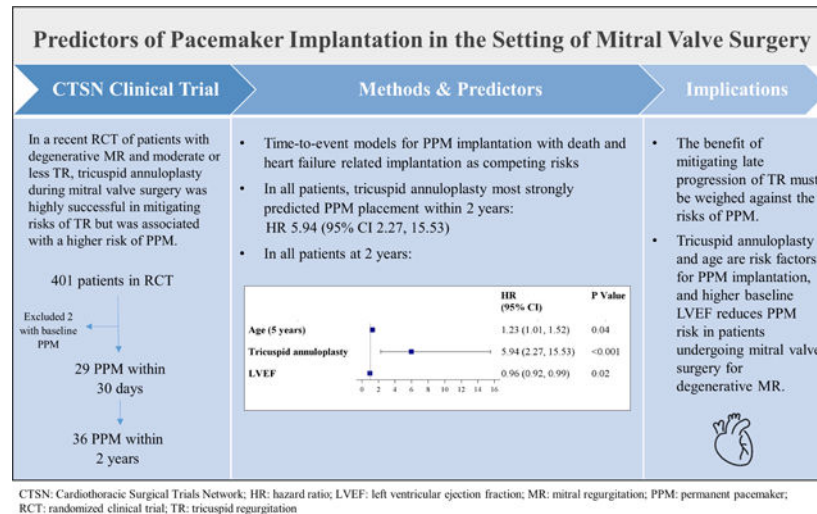
Objectives: In a recent trial, tricuspid annuloplasty (TA) during mitral surgery (MVS) for degenerative mitral regurgitation (MR) and moderate or less tricuspid regurgitation (TR) reduced the composite rate of death, re-operation for TR, or TR progression at 2 years. However, this benefit was counterbalanced by an increase in permanent pacemakers (PPM). Here, we analyze the timing, indications, and risk factors for these implantations.

Methods: We randomized 401 patients (MVS alone=203; MVS+TA=198). Potential risk factors for PPMs were assessed using multivariable time-to-event models with death and PPM implantation for heart failure indications as competing risks.

Results: A PPM was implanted in 36 patients (9.6%; 95 CI 6.8%–13.0%) within 2 years of randomization, with 30/187 (16.0%) in the MVS+TA and 6/188 (3.2%) in the MVS groups (RR 5.08; 95% CI 2.16–11.94; $p<0.001$). The majority (29/36, 80.6%) of implants occurred within 30 days postoperatively. Independent risk factors for PPMs within 2 years were TA (HR 5.94; 95% CI 2.27–15.53; $p<0.001$), increasing age (5 years, HR 1.23; 95% CI 1.01–1.52; $p=0.04$), and LVEF (HR 0.96; 95% CI 0.92–0.99; $p=0.02$). In the subset of TA recipients ($n=197$), age (5 years, HR 1.05; 95% CI 1.00–1.10; $p=0.04$) and LVEF (HR 0.95; 95% CI 0.91–0.99; $p=0.01$) were associated with PPM within 2 years.

Conclusions: Concomitant TA, age, and baseline LVEF were risk factors for PPM implantation in patients undergoing MV surgery for degenerative MR. While TA was effective in preventing progression of TR, innovation is needed to identify ways to decrease PPM rates.

Graphical Abstract



Keywords

pacemaker; mitral valve regurgitation; tricuspid annuloplasty; tricuspid regurgitation

INTRODUCTION

Tricuspid regurgitation (TR) is frequently encountered in the setting of severe degenerative mitral regurgitation (MR). A recent randomized trial from the Cardiothoracic Surgical Trials Network (CTSN) demonstrated that patients with moderate TR or less than moderate TR with annular dilation receiving tricuspid annuloplasty (TA) at the time of mitral valve surgery (MVS) for degenerative MR had a significantly lower 2-year rate of a composite endpoint of death, reoperation for TR, and progression of TR compared with patients undergoing MVS alone (3.9% vs. 10.2%; $p=0.02$) (1). This difference was driven by a substantially lower rate of progression of TR among patients assigned to TA. At 2 years, there was no discernable clinical impact of TA on mortality, symptoms, or rehospitalization rates. However, there was a substantially higher rate of permanent pacemaker (PPM) implantation in patients undergoing concomitant TA procedures.

Cardiac pacing, although a potentially lifesaving measure in the treatment of conduction disorders, may increase the risk of long-term adverse outcomes, including thrombosis, infection, pacing-induced ventricular dysfunction, and tricuspid valve dysfunction (2). Understanding of the risks of cardiac pacing and their long-term health implications is mostly derived from observational studies. However, there is great variation in the estimates of the incidence of PPM implantation (approximately 2.5%–15%) and their associated complication rates after mitral valve and tricuspid valve surgery (3, 4, 5). These rates are even higher if patients receive concomitant surgical atrial fibrillation (AF) ablation at the time of valve surgery (6, 7).

The CTSN tricuspid trial was designed to follow the progression of TR and related clinical endpoints in patients who had moderate or less TR and underwent mitral valve surgery with or without tricuspid annuloplasty for 5 years after randomization. Concomitant TA was

found to be successful in preventing progression of TR, but resulted in an increased need for PPM implantation. The objectives of this analysis were to evaluate the incidence, timing, mechanism, and risk factors related to PPM implantation among patients randomized in the CTSN tricuspid trial.

METHODS

Study Population and Interventions

The study population included 401 patients enrolled in a randomized trial evaluating the addition of concomitant TA in patients with moderate or less TR undergoing mitral valve surgery for severe degenerative disease. The primary objective of the trial was to evaluate the safety and effectiveness of concomitant TA; the latter was defined as the composite of death, re-operation for TR, and progression of TR from baseline by 2 grades or presence of severe TR at 2 years, with imputation for missing data. In this analysis, two patients were excluded because of baseline PPM implantation.

The target population comprised patients with either moderate TR or none/trace or mild TR with tricuspid annular dilation (≥ 40 mm or index: $\geq 21\text{mm}/\text{M}^2$ BSA). TR was assessed by transthoracic 2D echocardiography and verified by the central echocardiographic core laboratory. Patients were randomized (1:1 ratio) to MVS alone or MVS with tricuspid valve annuloplasty (MVS+TA). All patients underwent MVS via sternotomy or right mini-thoracotomy. The techniques of reparative mitral valve surgery, including suture placement and type of annuloplasty ring or valve, were at surgeon discretion. Recommendations for placing sutures from 9 to 6 o'clock (clockwise), and other aspects of surgical technique, were reviewed in an instructional video for all surgeon investigators (see video, supplemental appendix). The protocol specified an approved rigid, incomplete, nonplanar, and undersized (26, 28, or 30 mm) tricuspid annuloplasty ring (1). Commercially available tricuspid annuloplasty rings include: Edwards MC3 tricuspid annuloplasty ring (model 4900); Medtronic Contour 3D tricuspid annuloplasty ring (model 690R); ATS TriAd Tricuspid Annuloplasty Ring (model 900SFC); Carpentier-Edwards Classic tricuspid annuloplasty ring (model 4500); and Carpentier-Edwards Physio Tricuspid ring (model 6200).

The trial was designed by CTSN (1) and conducted in 34 centers (out of 39 registered) in the U.S., Canada, and Germany with central coordination, independent echocardiogram adjudication, independent clinical and adverse event adjudication, and an independent Data and Safety Monitoring Board (DSMB). Eligible surgeons needed to have performed at least 10 mitral valve operations and 5 tricuspid valve operations per year over the previous 2 year period. Institutional review boards at participating centers approved the protocol, and all patients signed a written informed consent. Complete inclusion and exclusion criteria have previously been reported (1).

Endpoints and Patient Assessments

The endpoint of interest in the current analysis was the incidence and timing of PPM implantation in the study population. PPM implantation decisions were made by the local

heart teams. Other outcomes of relevance to this secondary analysis include length of stay (LOS) for the index hospitalization and all-cause mortality at 2 years. Patients were evaluated at periodic intervals over 24 months after randomization and adverse events and mortality were collected on an event-driven basis. Survival will be evaluated for an additional 36 months

Statistical Analysis

Baseline characteristics and endpoints were compared between those who did and did not receive a permanent pacemaker implant during the entire follow-up period of 2 years and within 30 days of surgery. Patients who withdrew, died, or were lost to follow-up were not included in the two-year population. Baseline and operative characteristics were evaluated using T-tests, Wilcoxon rank sum tests, Chi-square tests, and Fisher's exact tests as appropriate. A Fine-Gray proportional sub-distribution hazards model was used to estimate the risk of PPM placement within 2 years and 30 days, with death and device implantation for heart failure indications as competing risks. All demographic, medical history, and operative characteristics in Tables 2 and 3 were evaluated as candidate risk factors and were included in the multivariable model if they were associated with the outcome at the 0.20 level of significance in univariable analyses. A stepwise backward selection approach was used to identify significant risk factors at the 0.10 level for the final multivariable model. Confidence intervals for pacemaker implantation were computed using the Clopper-Pearson exact method.

To evaluate the association of PPM with post-operative length of stay of the index hospitalization, the impairment of cardiac impulse formation or conduction abnormality adverse event that led to the placement of PPM was treated as a time-dependent variable in a Fine and Gray model with time to hospital discharge from surgery as the endpoint of interest and in-hospital death as a competing risk. Hazard ratios (HR) less than 1 corresponded to longer length of stay (i.e., a lower probability of being discharged early). All analyses were conducted using SAS version 9.4.

RESULTS

Incidence, Timing, and Indication for PPM Implantation

A total of 401 patients with severe degenerative MR and moderate or less TR (with annular dilation) were randomized to receive MVS with (n=198) or without TA (n=203). Two patients were excluded from this analysis due to a PPM at baseline (see CONSORT diagram, Figure E1). A PPM was implanted in 9.6% (95% confidence interval [CI] 6.8%–13.0%; n=36/375) of patients within 2 years of randomization, with a PPM incidence of 16.0% (n=30/187) among those randomized to MVS+TA and an incidence of 3.2% (n=6/188) among those randomized to MVS alone (RR 5.08; 95% CI 2.16–11.94; p<0.001). The overall incidence of PPM implantation at 30 days was 7.3% (95% CI 4.9%–10.3%; n=29/399), with a frequency of 11.6% (n=23/198) in the MVS+TA group versus 3.0% (n=6/201) in the MVS alone group (RR 3.91; 95% CI 1.63–9.40; p=0.002). The majority of PPM implants (80.6%; 29/36) occurred within 30 days of surgery (Figure 1). After 30 days, there were an additional 7 PPMs implanted (last implantation occurring at 611 days),

all of which occurred among patients who had received a concomitant TA. The median day of PPM implantation was 7.0 days (interquartile range [IQR] 6.0, 14.0).

Among the 36 PPMs implanted during the 2-year follow-up period, the most common indication was complete or high degree AV block (52.8%), followed by sinus node dysfunction (30.6%), and atrial fibrillation with high degree AV block (16.7%; Table 1). Table 1 also depicts the indications for implantation stratified by early and late implantation (within and after 30 days post-op). Atrial fibrillation with high degree AV block was not an indication for late PPM implantation.

Risk Factors for PPM Implantation within 30 Days and 2 Years

All Patients—The baseline and operative characteristics for all patients undergoing MVS, regardless of whether or not they had tricuspid valve surgery, and stratified by receipt of a PPM within 30 days and 2 years of index surgery, are depicted in Table 2. In univariate analyses, only TA (HR 4.07; 95% CI 1.67–9.93; $p=0.002$) was significantly associated with risk of PPM implantation within 30 days of surgery. Additional candidate risk factors for inclusion in the multivariable model were age (5 years; HR 1.18; 95% CI 0.96–1.44; $p=0.13$), history of AF (HR 1.61; 95% CI 0.78–3.34; $p=0.20$), concomitant maze (HR 1.76; 95% CI 0.84–3.71; $p=0.14$), and baseline LVEF (HR 0.97; 95% CI 0.94–1.01; $p=0.15$). Only two risk factors remained in the final multivariable model: age in increments of 5 years (HR 1.20; 95% CI 0.97–1.47; $p=0.09$) and TA (HR 4.24; 95% CI 1.73–10.41; $p=0.002$; Figure 2A).

There was a low incidence of PPM implantation between 30 days and 2 years confined to the group undergoing TA. Those receiving a PPM at 2 years were older (70.4 ± 9.9 vs 66.7 ± 10.2 years; $p=0.03$), more often had a history of atrial fibrillation (61.1% vs. 41.6%; $p=0.02$), and had lower LVEF at baseline (62.0 ± 6.8 vs 64.4 ± 7.4 ; $p=0.02$; Table 2). PPM recipients were more likely to have TA than those who did not receive a PPM (83.3% vs. 46.6%; $p<0.001$). Moreover, among patients receiving a PPM, 38.9% had a concomitant maze procedure as compared to 24.8% in the group who did not receive a PPM ($p=0.07$). In univariate analyses, TA (HR 5.37; 95% CI 2.24–12.89; $p<0.001$), LVEF (HR 0.96; 95% CI 0.93–0.99; $p=0.03$), and history of AF (HR 2.07; CI 1.06–4.02; $p=0.03$) were significantly associated with PPM implantation. Other candidate risk factors examined were age (5 years; HR 1.18; 95% CI 0.98–1.42; $p=0.07$), tricuspid annulus dimension (5 mm; HR 1.05, 95% CI 0.99–1.12; $p=0.12$), and concomitant maze (HR 1.84; 95% CI 0.95–3.59; $p=0.07$). The final multivariable model contained three risk factors, all significantly associated with PPM: age (5 years, HR 1.23, 95% CI 1.01–1.52; $p=0.04$), TA (HR 5.94; 95% CI 2.27–15.53; $p<0.001$), and LVEF (HR 0.96; 95% CI 0.92–0.99; $p=0.02$; Figure 2B).

Patients Undergoing Tricuspid Annuloplasty—In the subset of patients who received concomitant TA ($n=197$), those who required PPM within 30 days, compared with those who did not require PPM, were older (mean age 70.9 ± 10.2 years vs. 66.0 ± 10.7 years; $p=0.02$) and had slightly larger baseline tricuspid annulus dimensions (43.7 ± 4.6 mm vs. 41.7 ± 4.5 mm; $p=0.03$; Table 3). Univariate analysis identified the following candidate risk factors for PPM within 30 days: age (5 years; HR 1.26; 95% CI 0.99–1.60; $p=0.06$),

MV replacement (HR 2.41; 95% CI 0.88–6.63; $p=0.09$), tricuspid annulus dimension (5 mm; HR 1.54; 95% CI 0.99–2.41; $p=0.05$), history of AF (HR 1.71; 95% CI 0.76–3.86; $p=0.20$), concomitant maze (HR 1.73; 95% CI 0.76–3.97; $p=0.19$), and baseline LVEF (HR 0.96; 95% CI 0.91–1.01; $p=0.11$). The multivariable model included age (5 years; HR 1.23; 95% CI 0.96–1.57; $p=0.10$), MV replacement (HR 2.20; 95% CI 0.92–5.27; $p=0.08$), and tricuspid annulus dimension (5 mm; HR 1.46; 95% CI 0.95–2.26; $p=0.09$), but none were statistically significant (Figure 3A).

Focusing on the 2-year time period following TA, PPM recipients were older (mean age 70.9 ± 9.9 years vs. 65.5 ± 10.8 years; $p=0.008$), had higher rates of prior atrial fibrillation (63.3% vs. 40.1%; $p=0.02$) and reduced LVEF (61.6 ± 6.8 vs 64.7 ± 6.9 ; $p=0.01$) compared to those who did not receive a PPM (Table 3). In terms of operative characteristics, patients receiving PPM had larger tricuspid annulus dimensions (43.8 ± 4.3 vs 41.6 ± 4.5 ; $p=0.006$). Univariate analysis identified the following candidate risk factors for PPM within 2 years: age (5 years; HR 1.26; 95% CI 1.03–1.55; $p=0.03$), baseline LVEF (HR 0.95; 95% CI 0.91–0.99; $p=0.02$), TV repair implant size (HR 1.11; 95% CI 0.96–1.29; $p=0.17$), tricuspid annulus dimension (5 mm; HR 1.61; 95% CI 1.10–2.34; $p=0.01$), history of AF (HR 2.30; 95% CI 1.10–4.81; $p=0.03$), and concomitant maze (HR 1.82; 95% CI 0.88–3.75; $p=0.11$). The multivariable model included age (5 years; HR 1.05; 95% CI 1.00–1.10; $p=0.04$) and baseline LVEF (HR 0.95; 95% CI 0.91–0.99; $p=0.01$; Figure 3B).

Length of Stay and Mortality

Nearly all (28/29) of the PPMs implanted early (within 30 days) occurred before index hospital discharge. Overall, the median index hospitalization length of stay (LOS) was 9 days for patients with a PPM implanted within 30 days and 8 days for patients without PPM. However, LOS varied by country; it was 2 days longer (7 vs 9 days) for PPM recipients in the US; 4 days longer (8 vs 12 days) in Canada; and the median LOS was no different in Germany at 12 days, where the longer LOS is shaped by health care system practices (Figure 4). Analyzing time to discharge for patients in North America, the need for PPM implantation decreased the probability of early discharge (HR 0.74; 95% CI 0.60–0.87; $p<0.001$), adjusting for age. At two years, there were no deaths among patients implanted with a PPM (0/33) compared to a 4.2% mortality (15/354) in patients without a PPM implanted within 2 years.

DISCUSSION

In the present analysis, we observed that 16% of patients who underwent mitral valve surgery plus TA experienced conduction abnormalities that led to PPM implantation within 2 years, compared to only 3% of patients who underwent mitral valve surgery alone. Among patients who received a PPM in the first 30 days after surgery, age and TA were the only independent risk factors. Extending the analysis period to 2 years, TA, age, and baseline LVEF were independently associated with PPM insertion, with TA being the dominant risk factor (Figure 5). The factors that were not significantly associated with the risk of PPM included the ratio of tricuspid annuloplasty implant to annulus size, concomitant maze,

preoperative AF, technique for tricuspid repair (beating heart versus arrested heart), and access to the mitral valve (transeptal versus biatrial).

Patients with moderate or less TR receiving TA at the time of MVS for degenerative MR had a significantly lower 2-year rate of a composite endpoint of death, reoperation for TR, and progression of TR compared with patients undergoing MVS alone (3.9% vs. 10.2%; $p=0.02$). Moreover, the incidence of moderate or severe TR over 2 years was 3.4% in patients undergoing mitral valve surgery with concomitant TA and 25.1% in patients undergoing mitral valve surgery alone (RR 0.13; 95% CI 0.06–0.30). However, this dramatic reduction in TR carried a markedly increased risk of PPM placement. Observational studies of patients undergoing mitral and tricuspid valve surgery report PPM rates as low as 2.4% at individual centers to 14.7% in the STS Adult Cardiac Surgery Database. (3,4) Chikwe and colleagues reported the lowest pacemaker rate in the literature with mitral and concomitant tricuspid surgery, but this experience was in a younger population (mean age 57 years compared to 67 years in the CTSN study) (1). Moreover, observational studies may not have the infrastructure in place to capture all long-term PPM insertions, which may not occur at the index surgery hospital. In the present secondary analysis of our randomized trial data, all patients were closely followed out to 2 years, with adjudication of all adverse events. As such, the higher risk of PPM implantations seen in the present analysis may reflect, in part, a more rigorous follow-up and event adjudication.

The trial protocol recommended using an incomplete rigid nonplanar ring, preferably between 26 and 30 mm in size. These sizes were chosen as other studies had reported effective reduction in TR with the use of size 26–30mm TA rigid rings in the setting of severe TR (8,9). While TA was successful at reducing TR progression, there have been concerns that the downsizing of the tricuspid annulus itself may be responsible for an increased risk of the need for cardiac pacing. In TA recipients, the average annuloplasty ring size was 29.0 ± 1.9 mm for men and 27.8 ± 1.6 mm for women. We examined three specific variables within the 30-day time frame relative to the tricuspid valve, including TV annular dimension, repair implant size, and annulus implant ratio, but none were independently associated with the need for cardiac pacing. Moreover, the absence of any ring dehiscence in the TA group is evidence against too much downsizing.

The choice between semi-rigid or flexible rings has been raised as a potential factor affecting the outcomes after TA (10). Of historical importance, McCarthy and colleagues reported high recurrence of TR with the use of flexible bands (11). While freedom from TR is, in general, more stable with the use of a rigid ring, it is possible that greater geometric reverse remodeling with a rigid ring, in the setting of aggressive downsizing, could influence risk of AV block. Indeed, previous investigators have demonstrated more tricuspid annular reverse remodeling, higher risk of high degree atrioventricular block and higher risk of ring dehiscence (nearly 10%) in patients receiving a rigid or semi-rigid tricuspid annuloplasty ring (12,13). We analyzed the relationship between rigid and non-rigid rings on the incidence of PPM implantation and found no difference in this small sample. At 30 days, 12.5% (19/152) of patients with a rigid ring had a PPM implantation, as compared to 8.9% (4/45) of patients with a non-rigid ring ($p=0.51$). Moreover, as mentioned,

we did not identify any dehiscence. Further studies will be needed to better understand the influence of ring type and size on PPM risk.

In terms of intraoperative factors, previous studies have made the case for performing TA with a beating heart instead of with an arrested heart. In a recent multicenter study, Russo and colleagues demonstrated less postoperative renal failure and superior freedom from death or reoperation at 6 years with beating heart tricuspid surgery (14). Although PPM risk was not reported, others have identified AV node injury occurring with beating heart tricuspid surgery. In our study, beating heart tricuspid repair was performed in only 14% of patients with a 3.8% incidence of PPM implantation, while arrested heart surgery had a pacemaker incidence of 12.9% ($p=0.21$). As mentioned, beating heart surgery or other intraoperative variables, mitral valve replacement, trans-septal versus bi-atrial approach to the tricuspid valve, sternotomy versus thoracotomy, and concomitant maze were all not significantly associated with risk of PPM.

It remains unclear if surgical technique played a role in PPM rates. In this pragmatic trial, 84 surgeons in 34 centers randomized patients. During site training sessions and investigator meetings, surgeons discussed best practice techniques of annular suture placement (supplemental appendix) and reviewed trial educational materials, including videos of TV repair and the method of sizing of annuloplasty rings to minimize variation among centers. To what extent high- versus low-volume surgeons and clinical sites affect rates of PPM implantation is a question of interest. However, we did not collect the overall MV and TV repair volume per surgeon and site over time. Registries, such as the STS database and other non-US registries, are better suited to address this important issue.

A distinctive aspect of our study of two-year outcomes was that it was based on a cohort of trial patients, who are being followed longitudinally out to 5 years. Later PPM implantations (between 30 days and 2 years), occurring in 7 patients, were only seen in those who received a concomitant TA. The mechanisms for this risk have not been established, but may be due to ongoing remodeling of the tricuspid annulus or inflammation related to the annuloplasty ring. Further studies are needed to better characterize the risk of late need for cardiac pacing as well as the need for ongoing pacing among patients who receive a PPM. Moreover, more insight into the long-term clinical consequences of PPMs is needed. PPMs have been associated with device malfunction, thrombosis, infection, recurrent/progressive TR, RV remodeling and reduced survival (2,10). In this study, PPM insertion was not shown to have any effect on 2-year survival, but did have an upfront cost of prolonged hospitalization, at least in North American patients (This was not observed in Germany where the health care system subscribes to a longer length of stay, which includes cardiac rehabilitation). Without any mortality events among PPM recipients, one cannot calculate a risk ratio for the effect of PPM on death. Five year follow-up of these patients may, therefore, provide important insights regarding impact on survival.

Study Limitations

This study has several limitations. First, the indications for and timing of PPM implantation was left to the discretion of local heart teams, which resulted in a substantial number of implants occurring within the first 7 days. However, none were inserted less than 4

days postoperatively. Second, the study did not assess long term pacemaker dependency and it has been suggested that conduction defects resulting in PPM insertion may resolve spontaneously weeks after surgery. Third, the protocol did not dictate the technique for tricuspid annuloplasty or mitral repair/replacement, but only offered recommendations. In this comparative effectiveness trial, specifics of surgical technique were left to the discretion of experienced surgeons capturing the variation seen in practice and real-world outcomes. Fourth, while ascertainment of prior history or evidence of atrial fibrillation was part of the data collection for this trial, prior conduction disease was not and therefore this could not be included among the potential risk factors evaluated in our multivariable modeling. Moreover, we did not collect details on the ablation techniques used. Finally, there were relatively few PPMs for multivariable modeling which limits our ability to identify risk factors and affects the precision of our estimates.

CONCLUSIONS

Addition of tricuspid annuloplasty for moderate or less TR with annular dilation at time of MV surgery for degenerative MR was associated with a 6-fold higher risk of PPM compared with MV surgery alone. Risk factors for PPM implantation within 2 years following surgery included age, LVEF, and TA. For any given patient, the benefit of mitigating late progression of TR must be weighed against the risks of PPM. Optimal clinical decision-making requires insights into the long-term consequences of progressive TR and the clinical implications of PPM implantation, and patients will be followed for 5 years. Finally, further research needs to delineate opportunities to reduce the risk of PPM implantations associated with this procedure.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Glossary of Abbreviations

CTSN	Cardiothoracic Surgical Trials Network
DSMB	Data and Safety Monitoring Board
HR	hazard ratio
IQR	interquartile range
LOS	length of stay
MR	Mitral regurgitation

MVS	mitral valve surgery
PPM	permanent pacemakers
QoL	quality of life
SD	standard deviation
SF-12	12-Item Short-Form Health Survey
STS	Society for Thoracic Surgeons
TA	Tricuspid annuloplasty
TR	Tricuspid regurgitation

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Central Message

Tricuspid annuloplasty, age, and ejection fraction are independent risk factors for permanent pacemaker implantation in patients undergoing surgery for degenerative mitral regurgitation.

Perspective Statement

A recent CTSN trial documented significantly less tricuspid regurgitation in patients undergoing tricuspid annuloplasty for moderate tricuspid regurgitation or dilated tricuspid annulus during surgery for degenerative mitral regurgitation with an increased risk for permanent pacemaker (PPM) implantation. Independent risk factors for PPM include age, LVEF, and tricuspid annuloplasty.

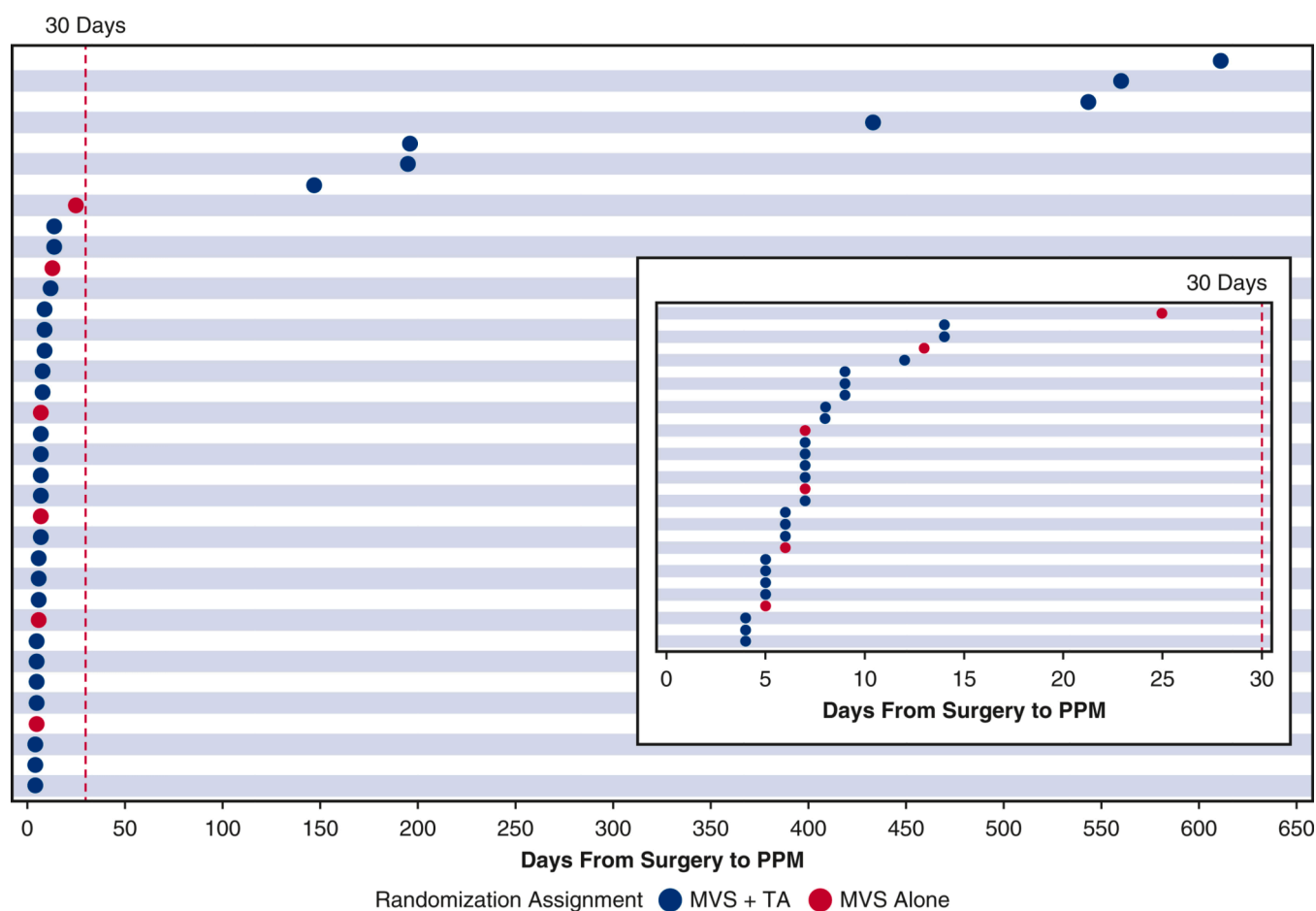
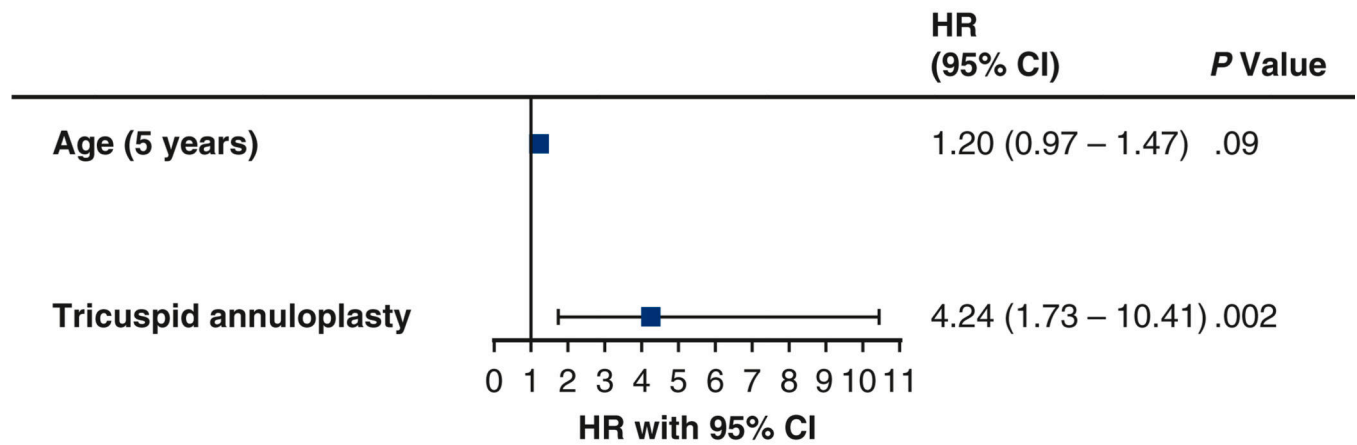
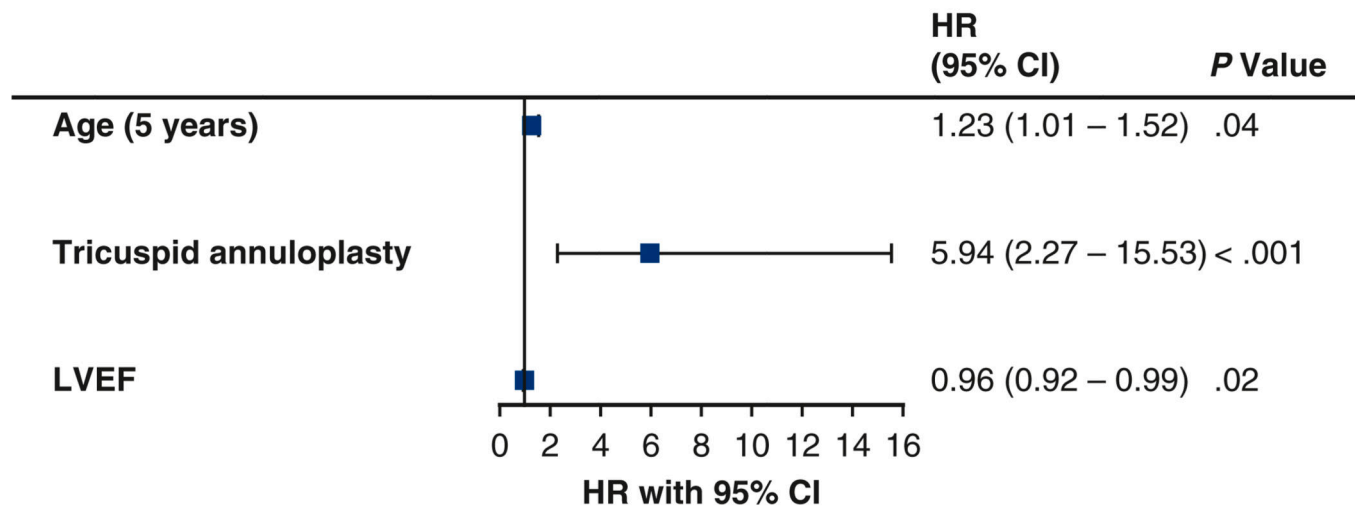


Figure 1.

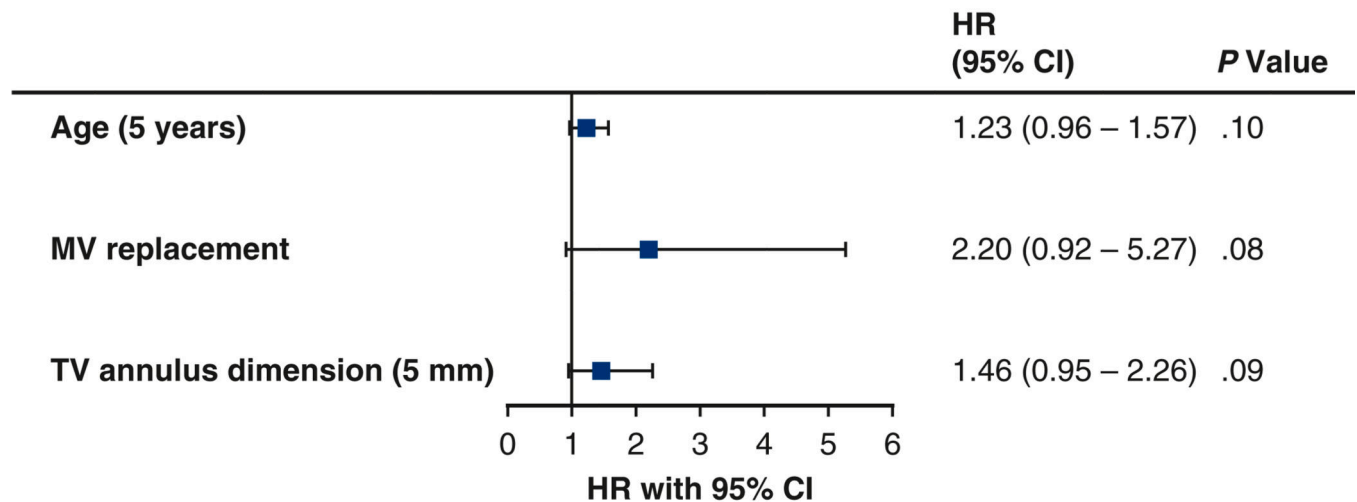
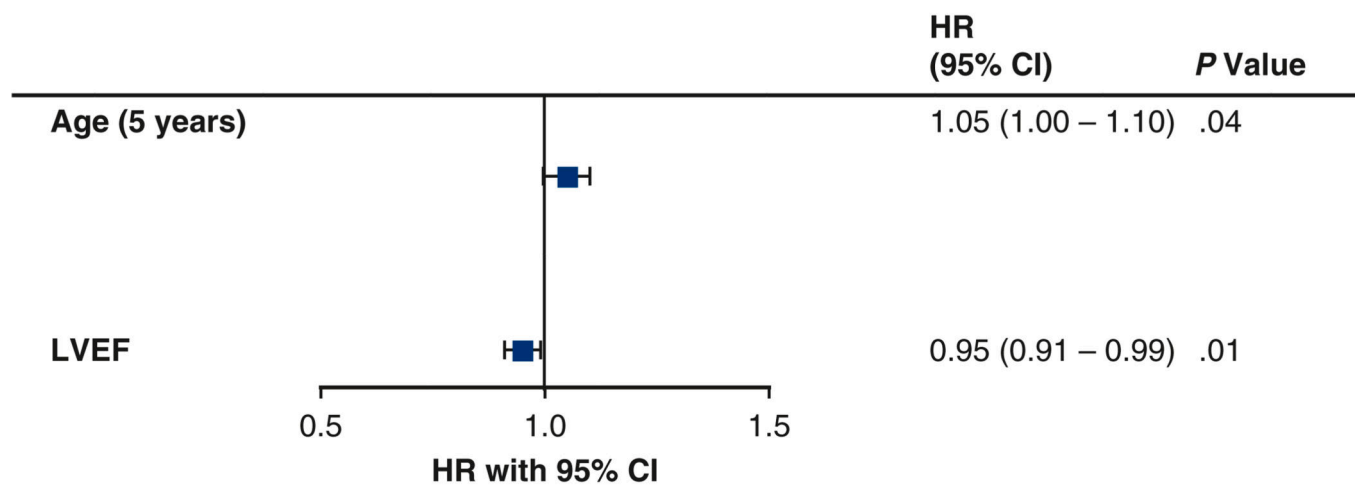
Timing of permanent pacemaker (PPM) implantation within 2 years in patients undergoing mitral valve surgery with or without concomitant tricuspid annuloplasty (N=36). Inset displays PPMs inserted within 30 days.

MVS, mitral valve surgery; *TA*, tricuspid annuloplasty

**A****B****Figure 2.**

Risk factors for permanent pacemaker implantation within 30 days (N=399) (A) and 2 years (N=382) (B) in patients undergoing mitral valve (MV) surgery with or without concomitant tricuspid annuloplasty. A Fine-Gray proportional sub-distribution hazards model was used, with death and device implantation for other indications as competing risks. A stepwise backward selection approach was used to identify significant risk factors at the 0.10 level for the final multivariable model. Seventeen patients are missing from the 2-year model due to missing left ventricular ejection fraction (LVEF).

HR, hazard ratio; CI, confidence interval.

**A****B****Figure 3.**

Risk factors for permanent pacemaker implantation within 30 days (N=195) (**A**) and 2 years (N=190) (**B**) in patients undergoing mitral valve (MV) surgery with concomitant tricuspid annuloplasty. A Fine-Gray proportional sub-distribution hazards model was used, with death and device implantation for other indications as competing risks. A stepwise backward selection approach was used to identify significant risk factors at the 0.10 level for the final multivariable model. Two patients with tricuspid annuloplasty are missing from the 30-day model due to missing tricuspid valve (TV) annulus dimension. Seven patients with tricuspid annuloplasty are missing from the 2-year model due to missing left ventricular ejection fraction (LVEF).

HR, hazard ratio; CI, confidence interval.

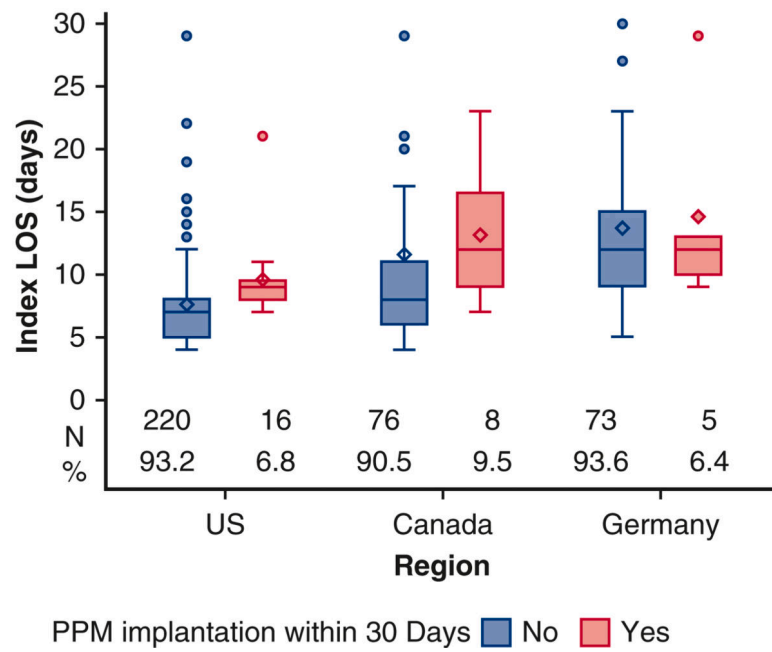


Figure 4.

Index hospitalization length of stay (LOS) in days by pacemaker implantation at 30 days and by region (n=398). One patient is missing due to missing length of stay. Data are shown as box-and-whisker plots. The lower and upper borders of the boxes represent the lower and upper quartiles. The middle horizontal lines represent the median and the diamonds represent the mean. The lower and upper whiskers represent the minimum and maximum values of non-outliers. Dots represent outliers.

PPM, permanent pacemaker.

Predictors of Pacemaker Implantation in the Setting of Mitral Valve Surgery

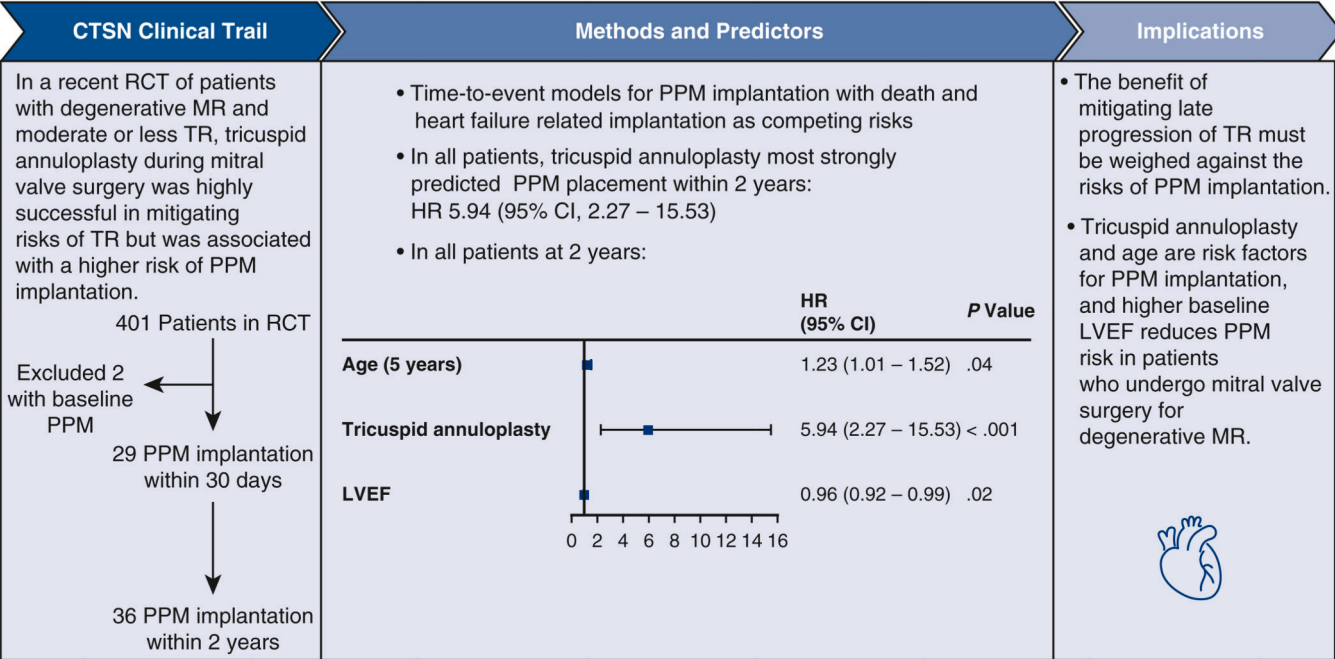
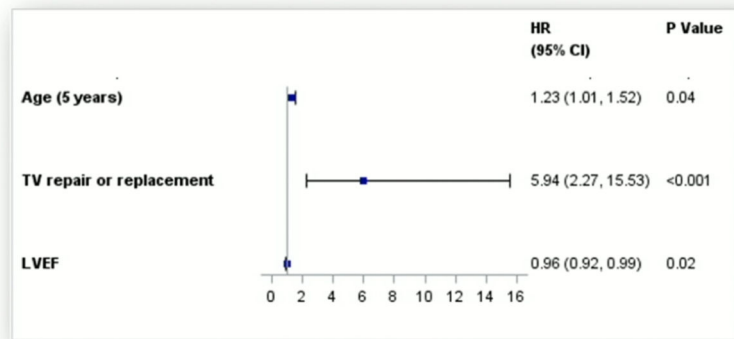


Figure 5. Predictors of pacemaker implantation after mitral valve surgery for degenerative mitral regurgitation and moderate or less tricuspid regurgitation.

Risk Factors for PPM at Two Years in All Patients

Univariable			Multivariable		
	HR (95% CI)	P value		HR (95% CI)	P value
Age (5 years)	1.18 (0.98, 1.42)	0.07	Age (5 years)	1.23 (1.01, 1.52)	0.04
TV repair or replacement	5.37 (2.24, 12.89)	<0.001	TV repair or replacement	5.94 (2.27, 15.53)	<0.001
LVEF	0.96 (0.93, 0.99)	0.03	LVEF	0.96 (0.92, 0.99)	0.02
TV annulus dimension – AP4 view (5 mm)	1.05 (0.99, 1.12)	0.12			
History of atrial fibrillation	2.07 (1.06, 4.02)	0.03			
Concomitant MAZE	1.84 (0.95, 3.59)	0.07			



Central Picture.

Risk factors for PPM implant within 2 years of MV surgery with and without TA (N=382).

Table 1.

Indication for pacemaker implantation within and after 30 days after index surgery.

	Total (N=36)		Within 30 Days (N=29)		After 30 Days (N=7)	
	N	%	N	%	N	%
Atrial fibrillation with high degree AV block	6	16.7	6	20.7	0	0.0
Complete or high degree AV block	19	52.8	16	55.2	3	42.9
Sinus node dysfunction	11	30.6	7	24.1	4	57.1

AV, atrioventricular

Baseline and operative characteristics by permanent pacemaker (PPM) implantation within 30 days (N=399) and 2 years (N=375) in patients undergoing mitral valve surgery with or without concomitant tricuspid annuloplasty.

Table 2.

	PPM Placement Within 30 Days			PPM Placement Within 2 Years*		
	No PPM (N=370)	PPM (N=29)	P value	No PPM (N=339)	PPM (N=36)	P value
Age, years	67.1 ± 10.2	70.4 ± 10.2	0.07	66.7 ± 10.2	70.4 ± 9.9	0.03
Male sex	276 (74.6)	22 (75.9)	0.88	255 (75.2)	28 (77.8)	0.73
BMI (kg/m ²)	26.5 ± 4.5	26.3 ± 4.2	0.95	26.6 ± 4.6	26.4 ± 4.0	0.90
Race			>0.99			0.84
American Indian or Alaska Native	1 (0.3)	0 (0.0)		1 (0.3)	0 (0.0)	
Asian	8 (2.3)	0 (0.0)		7 (2.1)	0 (0.0)	
Black or African American	10 (2.8)	0 (0.0)		10 (3.1)	0 (0.0)	
Native Hawaiian or Other Pacific Islander	1 (0.3)	0 (0.0)		1 (0.3)	0 (0.0)	
White	335 (94.4)	29 (100.0)		307 (94.2)	34 (100.0)	
Hispanic or Latino	6 (1.7)	0 (0.0)	>0.99	5 (1.5)	1 (2.9)	0.46
Medical History						
Atrial Fibrillation	159 (43.0)	16 (55.2)	0.20	141 (41.6)	22 (61.1)	0.02
Paroxysmal	80 (21.6)	6 (20.7)		71 (20.9)	9 (25.0)	
Persistent	79 (21.4)	10 (34.5)		70 (20.6)	13 (36.1)	
Ventricular arrhythmia	29 (7.8)	1 (3.4)	0.71	25 (7.4)	3 (8.3)	0.74
Valve repair**	6 (1.6)	0 (0.0)	>0.99	6 (1.8)	0 (0.0)	>0.99
Valve replacement†	4 (1.1)	0 (0.0)	>0.99	4 (1.2)	0 (0.0)	>0.99
Stroke or TIA	17 (4.6)	0 (0.0)	0.62	15 (4.4)	1 (2.8)	>0.99
Diabetes	25 (6.8)	1 (3.4)	0.71	22 (6.5)	1 (2.8)	0.71
Hypertension	216 (58.4)	18 (62.1)	0.70	197 (58.1)	24 (66.7)	0.32
NYHA Class III/IV	112 (30.4)	8 (27.6)	0.75	100 (29.6)	11 (30.6)	0.90
LVEF	64.3 ± 7.3	62.6 ± 6.6	0.07	64.4 ± 7.4	62.0 ± 6.8	0.02
Index Procedure						
TV procedure			0.002			<0.001

	PPM Placement Within 30 Days			PPM Placement Within 2 Years*		
	No PPM (N=370)	PPM (N=29)	P value	No PPM (N=339)	PPM (N=36)	P value
None	195 (52.7)	6 (20.7)		181 (53.4)	6 (16.7)	
Repair	174 (47.0)	23 (79.3)		157 (46.3)	30 (83.3)	
Replacement	1 (0.3)	0 (0.0)		1 (0.3)	0 (0.0)	
MV procedure			0.20			0.40
Repair	334 (90.3)	24 (82.8)		305 (90.0)	31 (86.1)	
Replacement	36 (9.7)	5 (17.2)		34 (10.0)	5 (13.9)	
Approach type			0.50			0.48
Stemotomy	193 (52.2)	17 (58.6)		177 (52.2)	21 (58.3)	
Thoracotomy	177 (47.8)	12 (41.4)		162 (47.8)	15 (41.7)	
Concomitant maze	94 (25.4)	11 (37.9)	0.14	84 (24.8)	14 (38.9)	0.07
Biatrial	44 (46.8)	5 (45.5)		40 (47.6)	7 (50.0)	
Left atrial	42 (44.7)	5 (45.5)		36 (42.9)	6 (42.9)	
Unknown	8 (8.5)	1 (9.1)		8 (9.5)	1 (7.1)	
Intraoperative amiodarone	54 (14.6)	2 (6.9)	0.40	48 (14.2)	3 (8.3)	0.45
OR time, min	347.4 ± 110.1	334.3 ± 110.6	0.60	345.3 (107.8)	348.3 ± 120.9	>0.99
CPB time, min	149.0 ± 67.0	152.6 ± 59.7	0.55	148.0 (66.3)	156.5 ± 64.2	0.37
TV annulus dimension – AP4 view (mm)	42.0 ± 4.6	42.9 ± 4.7	0.21	42.0 ± 4.6	43.2 ± 4.5	0.07

* Patients who withdrew, died, or were lost to follow-up are not included in the two-year population.

*** The six previous valve repairs were MV surgical.

† The four previous valve replacements consist of three AVR bioprosthetics and one MVR mechanical.

Chi-square and Fisher's exact tests were used for categorical variables, as appropriate.

Categorical measures are presented as number of patients and (%). If the denominator is not equal to the group sample size, data are presented as number of patients/the number observed (%). Continuous measures are presented as mean ± standard deviation.

Wilcoxon rank sum tests were used for continuous variables.

BMI, body mass index; *TIA*, transient ischemic attack; *NYHA*, New York Heart Association; *LVEF*, left ventricular ejection fraction; *TV*, tricuspid valve; *MV*, mitral valve; *OR*, operating room; *CPB*, cardiopulmonary bypass.

Table 3.

Baseline and operative characteristics among patients who did or did not receive a permanent pacemaker (PPM) implantation within 30 days (N=197) and 2 years (N=187) of mitral valve surgery with concomitant tricuspid annuloplasty.

	PPM Placement Within 30 Days			PPM Placement Within 2 Years*		
	No PPM (N=174)	PPM (N=23)	P value	No PPM (N=157)	PPM (N=30)	P value
Age, years	66.0 ± 10.7	70.9 ± 10.2	0.02	65.5 ± 10.8	70.9 ± 9.9	0.008
Male sex	128 (73.6)	18 (78.3)	0.63	116 (73.9)	24 (80.0)	0.48
BMI (kg/m ²)	26.7 ± 4.5	26.3 ± 4.1	0.79	26.8 ± 4.5	26.4 ± 3.8	0.91
Race			>0.99			>0.99
Asian	3 (1.8)	0 (0.0)		2 (1.3)	0 (0.0)	
Black or African American	3 (1.8)	0 (0.0)		3 (2.0)	0 (0.0)	
Native Hawaiian or other Pacific Islander	1 (0.6)	0 (0.0)		1 (0.7)	0 (0.0)	
White	158 (95.8)	23 (100.0)		144 (96.0)	28 (100.0)	
Hispanic or Latino	5 (3.0)	0 (0.0)	>0.99	4 (2.6)	1 (3.3)	>0.99
Medical History						
Atrial Fibrillation	74 (42.5)	13 (56.5)	0.20	63 (40.1)	19 (63.3)	0.02
Paroxysmal	36 (20.7)	4 (17.4)		29 (18.5)	7 (23.3)	
Persistent	38 (21.8)	9 (39.1)		34 (21.7)	12 (40.0)	
Ventricular arrhythmia	16 (9.2)	1 (4.3)	0.70	13 (8.3)	3 (10.0)	0.72
Valve repair**	4 (2.3)	0 (0.0)	>0.99	4 (2.5)	0 (0.0)	>0.99
Valve replacement†	1 (0.6)	0 (0.0)	>0.99	1 (0.6)	0 (0.0)	>0.99
Stroke or TIA	9 (5.2)	0 (0.0)	0.60	7 (4.5)	1 (3.3)	>0.99
Diabetes	14 (8.0)	1 (4.3)	>0.99	13 (8.3)	1 (3.3)	0.70
Hypertension	97 (55.7)	14 (60.9)	0.64	86 (54.8)	20 (66.7)	0.23
NYHA Class III/IV	48 (27.7)	5 (21.7)	0.54	41 (26.3)	8 (26.7)	0.97
LVEF	64.4 ± 6.9	62.1 ± 6.6	0.06	64.7 (6.9)	61.6 (6.8)	0.01
Index Procedure						
MV procedure			0.10			0.29
Repair	162 (93.1)	19 (82.6)		145 (92.4)	26 (86.7)	

	PPM Placement Within 30 Days			PPM Placement Within 2 Years*		
	No PPM (N=174)	PPM (N=23)	P value	No PPM (N=157)	PPM (N=30)	P value
Replacement	12 (6.9)	4 (17.4)		12 (7.6)	4 (13.3)	
TV repair performed			0.21			0.77
Beating heart	26 (14.9)	1 (4.3)		22 (14.0)	3 (10.0)	
Arrested	148 (85.1)	22 (95.7)		135 (86.0)	27 (90.0)	
Approach to TV repair			0.70			0.32
Trans-septal	18 (10.3)	1 (4.3)		17 (10.8)	1 (3.3)	
Bi-atriotomy	156 (89.7)	22 (95.7)		140 (89.2)	29 (96.7)	
Approach type			0.50			0.47
Stemotomy	93 (53.4)	14 (60.9)		83 (52.9)	18 (60.0)	
Thoracotomy	81 (46.6)	9 (39.1)		74 (47.1)	12 (40.0)	
Concomitant maze	46 (26.4)	9 (39.1)	0.20	41 (26.1)	12 (40.0)	0.12
Biatrial	25 (54.3)	5 (55.6)		22 (53.7)	7 (58.3)	
Left atrial	16 (34.8)	3 (33.3)		14 (34.1)	4 (33.3)	
Unknown	5 (10.9)	1 (11.1)		5 (12.2)	1 (8.3)	
Intraoperative amiodarone	19 (10.9)	2 (8.7)	>0.99	17 (10.8)	3 (10.0)	>0.99
OR time, min	367.9 ± 109.4	343.9 ± 103.0	0.30	364.3 ± 103.5	358.4 ± 116.6	0.59
CPB time, min	167.1 ± 70.9	158.8 ± 57.5	0.89	166.6 ± 71.3	162.0 ± 63.3	0.95
TV annulus dimension – AP4 view (mm)	41.7 ± 4.5	43.7 ± 4.6	0.03	41.6 ± 4.5	43.8 ± 4.3	0.006
TV repair implant size (mm)	28.7 ± 1.9	29.0 ± 1.7	0.28	28.6 ± 1.9	29.1 ± 1.7	0.13
TV repair implant type			0.51			0.30
Rigid non-planar or rigid planar rings	133 (76.4)	19 (82.6)		117 (74.5)	25 (83.3)	
Non-rigid rings	41 (23.6)	4 (17.4)		40 (25.5)	5 (16.7)	
TV annulus implant ratio	1.5 ± 0.2	1.5 ± 0.2	0.16	1.5 ± 0.2	1.5 ± 0.2	0.11

* Patients who withdrew, died, or were lost to follow-up are not included in the two-year population.

** The four previous valve repairs were MV surgical.

† The one previous valve replacements was an AVR bioprosthetic.

Categorical measures are presented as number of patients and (%). If the denominator is not equal to the group sample size, data are presented as number of patients/the number observed (%). Continuous measures are presented as mean ± standard deviation.

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Chi-square and Fisher’s exact tests were used for categorical variables, as appropriate.
T-tests and Wilcoxon rank sum tests were used for continuous variables, as appropriate.
BMI, body mass index; *TIA*, transient ischemic attack; *NYHA*, New York Heart Association; *LVEF*, left ventricular ejection fraction; *TV*, tricuspid valve; *MV*, mitral valve; *OR*, operating room; *CPB*, cardiopulmonary bypass.